12.14 Mixed Strategies

WORKSHEET

The question

STEP I 1989, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q16

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Write down a table of B's winnings for each pair of guesses (B's guess down the side, A's across the top).

STAGE 2 Against random guessing

Ask yourself If A guesses uniformly at random, what is B's expected gain when B says j? (Average along row j of the table.)

STAGE 3 The general case

Ask yourself Now A plays j with probability a_j. Collect B's expected winnings by A's guess: what is the coefficient of a_1 (the average of column 1 with weights b_j)?

Ask yourself Can you make all three zero? (Three equations, but they are not independent; why not?)

Try this next

1. **A's best reply.** If B uses $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$, show that every strategy of A gives expected winnings 0. If instead B uses $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, find A's best reply.

2. **Rock, paper, scissors.** Write the payoff matrix for rock–paper–scissors and find the optimal strategy. Why is it uniform here but not in the question?

3. **Uniqueness.** Show that $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ is the *only* strategy for B that guarantees not losing. (*Hint:* it must make X, Y, Z all non-negative; use $X + 2Y + Z = 0$.)

4. **A modified game.** If the penalty for differing by 2 is £3 instead of £2, find the optimal strategy.

Further reading

- **Ken Binmore, *Playing for Real*** (Oxford University Press). A thorough but friendly introduction to game theory.
- **Avinash Dixit & Barry Nalebuff, *The Art of Strategy*** (Norton). Game theory with examples from sport and business.

- **William Poundstone, *Prisoner's Dilemma*** (Anchor). Von Neumann, game theory and the Cold War.

The full guided solution, and what the question leads to at university, is in Section 12.14:
https://beyondstep.keithmcnulty.org/ch12/mixed-strategies.html